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Bachelor of Liberal Arts & Sciences

ENGDATA302 Robotics

Project Report

Inverse Kinematics for Human Fingers;

Iterative Solver Methods

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1 Introduction

The human hand consists of three main parts: the writs, palm, and five distinct fingers. Each finger is a structure of multiple joints and boney structures. These boney structures are also called phalanges. All fingers have the following three phalanges (except for the thumb which has only two phalanges):

1. The proximal phalanx is the bone closest to the palm and is connected with it by a so-called metacarpal.
2. The intermediate phalanx is the phalanx right after the proximal phalanx looking from the palm.
3. Lastly, the distal phalanx is the bone furthest away in the finger looking from the palm.

Tab. 1 shows the average lengths of these phalanges for the index, middle, ring, and little finger respectively.

Table 1: Average lengths of the phalanges for each finger (except the thumb) [1].

Finger	**Proximal** (p) **(mm)**	**Intermediate** (i) **(mm)**	**Distal** (d) **(mm)**
Index	39.8 (pi)	22.4 (ip)	15.8 (di)
Middle	44.6 (pm)	26.3 (im)	17.4 (dm)
Ring	41.4 (pr)	25.7 (ir)	17.3 (dr)
Little	32.7 (pl)	18.1 (il)	16.0 (dl)

Furthermore, each finger has three primarily revolute joints which let the fingers bend and move.

1. The metacarpophalangeal joint connects the proximal phalanx to the metacarpal in the palm. For the kinematic model of the fingers, it will be assumed that this joint is located at the origin of the coordinate frame.
2. The proximal interphalangeal joint connects the proximal phalanx to the intermediate phalanx.
3. The distal interphalangeal joint connects the intermediate phalanx to the distal phalanx.

To simplify matters, the thumb is excluded from the current analysis and all joint axes are parallel to the z-axis, meaning that the revolute joints have one degree of freedom and all motion takes place in the x, y-plane.

The angles of the metacarpophalangeal, proximal, and distal interphalangeal joints are denoted as θ_m, θ_p, and θ_d respectively. θ_m is the angle between the positive x-axis and proximal phalanx, θ_p is the angle between the intermediate phalanx and the extension of the proximal phalanx, and θ_d is the angle between the distal phalanx and the extension of the intermediate phalanx. All angles are measured in the counter-clockwise direction. For the rest of this report, the joint angle of a specific finger is denoted by its subscript, for example, θ_{mi} is the θ_m angle for the index finger.

Lastly, the model that will be considered includes a single unbounded object (O) which is defined as

$$O = \left\{(x, y) \in \mathbb{R}^2 | y + 18 \leq 0\right\}. \tag{1}$$

I.e., all space that is under the line of $y = -18$ is in the object.

The primary objective of this research project will be to study the kinematics of the human fingers. The aim is to put the fingertip on a specific target location or move along a pre-determined trajectory. To do this, the inverse kinematics of the model needs to be determined. Usually, the best method to do this is to look at the geometrical relations within the model to find an analytical solution. However, if not enough relations are given, an analytical solution cannot be found and numerical or iterative methods need to be used instead. As it turns out, these methods can sometimes give problems or a non-sufficient solution. It will be the goal of this project to optimize an iterative method as far as possible, such that a sufficient solution will be returned no matter the desired target positions (provided that these positions are within the reach of the finger).

2 Forward Kinematics

The first step of this project is to analyze the human fingers' kinematic model by looking at the forward kinematics. The forward kinematics determine the position of the fingertip as a function of the joint angles. Usually, the Denavit-Hartenberg algorithm is used to define the forward kinematics, but since the model is placed in a 2D scenario, the necessary equations can be determined using simple trigonometry. Fig. 1 shows the standard kinematic model for a human finger.

From this model, it can be seen that the x and y position of the finger tip are determined as

$$\begin{aligned} x_m &= m\cos(\theta_m) \\ x_p &= p\cos(\theta_m + \theta_p) \\ x_d &= d\cos(\theta_m + \theta_p + \theta_d) \\ x = x_m + x_p + x_d &= m\cos(\theta_m) + p\cos(\theta_m + \theta_p) + d\cos(\theta_m + \theta_p + \theta_d) \end{aligned} \tag{2}$$

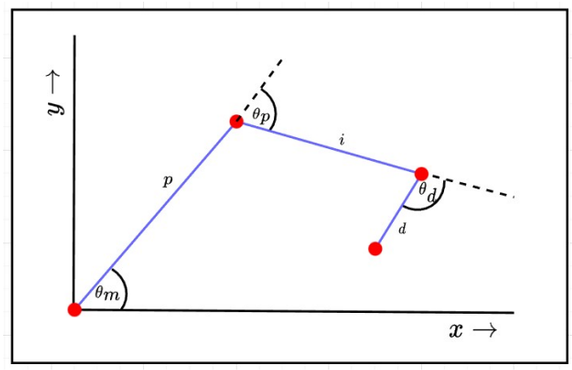


Figure 1: The kinematic model of a human finger.

and

$$\begin{aligned} y_m &= m\sin(\theta_m) \\ y_p &= p\sin(\theta_m+\theta_p) \\ y_d &= d\sin(\theta_m+\theta_p+\theta_d) \\ y = y_m + y_p + y_d &= m\sin(\theta_m) + p\sin(\theta_m+\theta_p) + d\sin(\theta_m+\theta_p+\theta_d), \end{aligned} \tag{3}$$

where $(x_m, y_m), (x_p, y_p)$, and (x_d, y_d) are the horizontal and vertical displacement of the proximal, intermediate, and distal phalanges respectively.

In the coming sections, the kinematics of the fingers will be studied more thoroughly by evaluating a number of interesting configurations of the joint angles that tell something about the boundary cases of the model.

2.1 Unconstrained Joint Angles

First, a rather simple model of the fingers is used for the analysis. The joint angles of the fingers are considered to be independent of each other and potential intersections of the phalanges are neglected. Furthermore, it is assumed that $-\frac{\pi}{3} \leq \theta_m \leq \frac{\pi}{3}$, and $-\frac{2\pi}{3} \leq \theta_p \leq 0$, and $-\frac{2\pi}{3} \leq \theta_p \leq 0$ in order to let the fingers behave more naturally.

2.1.1 Stretched Finger / Reachable Space

The reachable space of the fingertips describe a circle with a radius that is equal to the length of the finger while it is completely stretched. This length is simply the sum of the lengths of the individual phalanges. The conditions for stretching the fingers are $\theta_p = \theta_d = 0$, while θ_m can have any value within the allowable values.

The statement above can be used to find the point that is furthest along the obstacle (but still within the reachable space of the finger). This point will set a ceiling on the possible x-coordinates along the obstacle. Consider the finger to be stretched and to touch the object. Eq. 3 will now simplify and can be used to find the required value for θ_m:

$$\begin{aligned} y &= m\sin\theta_m + p\sin(\theta_m+0) + d\sin(\theta_m+0+0) \\ y &= (m+p+d)\sin\theta_m = -18 \\ \longrightarrow \theta_m &= \arcsin\left(\frac{-18}{m+p+d}\right). \end{aligned} \tag{4}$$

Inserting the values for m, p, and d for each finger as given in Tab. 1, will return the values for θ_m

$$\begin{aligned} \theta_{mi} &= \arcsin\left(\frac{-18}{39.8+22.4+15.8}\right) \approx -13.34^\circ \\ \theta_{mm} &= \arcsin\left(\frac{-18}{44.6+26.3+17.4}\right) \approx -11.76^\circ \\ \theta_{mr} &= \arcsin\left(\frac{-18}{41.4+25.7+17.3}\right) \approx -12.31^\circ \\ \theta_{ml} &= \arcsin\left(\frac{-18}{32.7+18.1+16.0}\right) \approx -15.63^\circ \end{aligned} \tag{5}$$

Fig. 2 shows the configuration of the fingers after the values for θ_m has been converted to an x, y position for the fingertip, and the phalanges, using Eq. 2 and 3.

2.1.2 Horizontal Proximal Phalanx

In this example, the proximal phalanx is set to be horizontal, implying that $\theta_m = 0$, and the challenge is to find the values for the other joint angles such that the fingertip touches the object. To make the analysis less difficult, only the case where the intermediate and distal phalanx are stretched is considered. Therefore,

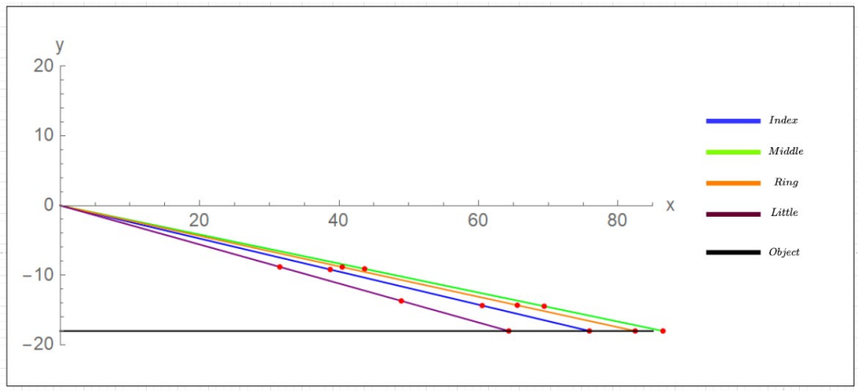


Figure 2: Fingertips placed at the point furthest along the object.

$\theta_d = 0$. Again, Eq. 3 will be simplified and can be used to find the values for the remaining joint angle:

$$\begin{aligned} y &= p\sin(0) + i\sin(0+\theta_p) + d\sin(0+\theta_p+0) = -18 \\ y &= (i+d)\sin(\theta_p) = -18 \\ \longrightarrow \theta_p &= \arcsin\left(\frac{-18}{i+d}\right) \end{aligned} \tag{6}$$

Inserting the values for i, and d, for each finger as given in Tab. 1, will return the values for θ_p of each finger:

$$\begin{aligned} \theta_{pi} &= \arcsin\left(\frac{-18}{39.8+22.4+15.8}\right) \approx -28.11^\circ \\ \theta_{pm} &= \arcsin\left(\frac{-18}{44.6+26.3+17.4}\right) \approx -24.32^\circ \\ \theta_{pr} &= \arcsin\left(\frac{-18}{41.4+25.7+17.3}\right) \approx -24.75^\circ \\ \theta_{pl} &= \arcsin\left(\frac{-18}{32.7+18.1+16.0}\right) \approx -31.86^\circ \end{aligned} . \tag{7}$$

Fig. 3 shows the configuration of the fingers after the values for θ_p has been converted to an x, y position for the fingertip, and the phalanges, using the forward kinematics.

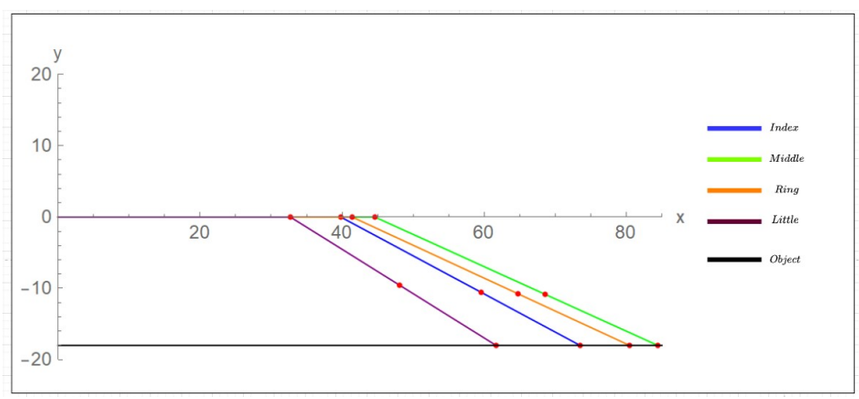


Figure 3: Fingertips placed on the object while their proximal phalanges are horizontal.

2.1.3 Vertical Distal Phalanx

Another interesting configuration is when the distal phalanx is placed horizontally, while the rest of the finger is stretched. As a consequence, $\theta_p = 0$ since the intermediate phalanx is not rotated relative to the proximal phalanx. Fig 4 gives a sketch of the current situation. From this sketch, it can be seen that Δy is the height at which the object is placed minus the length of the distal phalanx:

$$\Delta y = -18 + d. \tag{8}$$

Therefore, θ_m can be computed using the trigonometrical relation of

$$\begin{aligned} \sin(\theta_m) &= \frac{\Delta y}{p+i} \\ \longrightarrow \theta_m &= \arcsin\left(\frac{\Delta y}{p+i}\right). \end{aligned} \tag{9}$$

The only angle left to be determined is θ_d. Imagine a horizontal line going through the distal joint. The angle between the intermediate phalanx and this line has the same magnitude as θ_m, because they are z-angles.

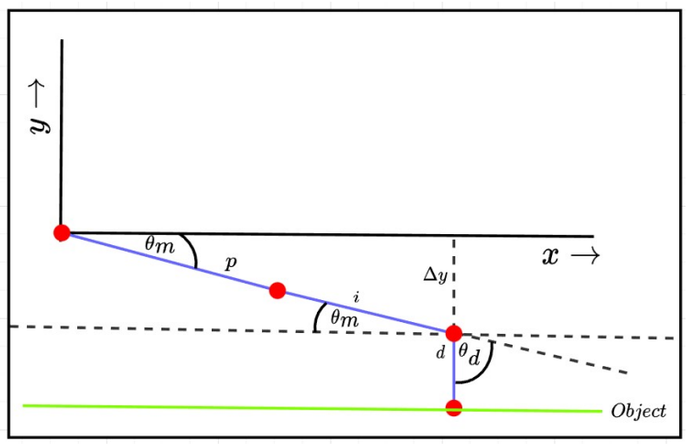


Figure 4: Overview of the kinematic model when the distal phalanx is vertical.

Therefore, the distal joint angle is determined as

$$\theta_d = -\frac{1}{2}\pi - \theta_m. \tag{10}$$

Using the values for p, i and d of Tab. 1, the values for θ_m of each finger are denoted as

$$\begin{aligned}
\theta_{mi} &= \arcsin\left(\frac{-18+15.8}{39.8+22.4}\right) \approx -0.035 \quad (-2.027^\circ) \\
\theta_{mm} &= \arcsin\left(\frac{-18+17.4}{44.6+26.3}\right) \approx -0.0085 \quad (-0.485^\circ) \\
\theta_{mr} &= \arcsin\left(\frac{-18+17.3}{41.4+25.7}\right) \approx -0.010 \quad (-0.5977^\circ) \\
\theta_{ml} &= \arcsin\left(\frac{-18+16.0}{32.7+18.1}\right) \approx -0.039 \quad (-2.26^\circ)
\end{aligned}, \tag{11}$$

and therefore the angles of θ_d are

$$\begin{aligned}
\theta_{di} &= -\frac{1}{2}\pi - (-0.035) \approx -1.54 \quad (-87.97^\circ) \\
\theta_{dm} &= -\frac{1}{2}\pi - (-0.0085) \approx -1.56 \quad (-89.52^\circ) \\
\theta_{dr} &= -\frac{1}{2}\pi - (-0.010) \approx -1.56 \quad (-89.40^\circ) \\
\theta_{dl} &= -\frac{1}{2}\pi - (-0.039) \approx -1.53 \quad (-87.74^\circ)
\end{aligned}. \tag{12}$$

Fig. 5 shows the configuration of the fingers after the values for θ_d have been converted to an x, y position for the fingertip, and the phalanges, using the equations of forward kinematics.

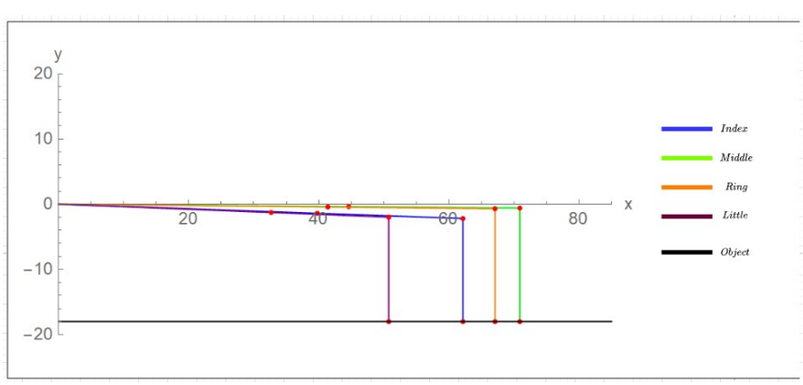


Figure 5: Fingertips placed on the object while the distal phalanges are vertical.

2.1.4 Horizontal Proximal Phalanx and Vertical Distal Phalanx

The next step would be to find the set of joint angles for which both the proximal phalanx is horizontal and the distal phalanx is vertical, while the fingertip still touches the object. Fig. 6 shows a sketch of this situation. In this case, θ_m is zero and the angle θ_p can be derived similarly as in Eq. 9, but now the length of p should be omitted:

$$\theta_p = \sin\left(\frac{\Delta y}{i}\right). \tag{13}$$

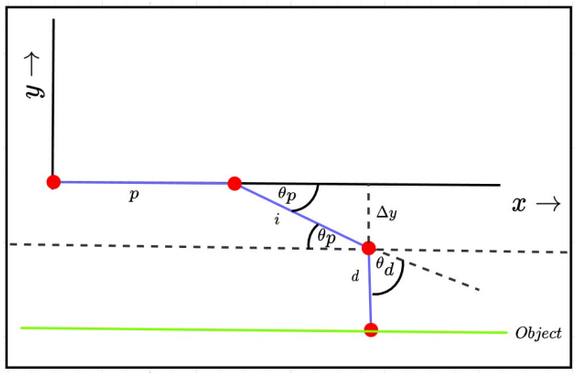


Figure 6: Sketch of the situation where the proximal phalanx is horizontal and the distal phalanx is vertical.

The derivation for the angle θ_d is in line with Eq. 10, but θ_m should now be replaced with θ_p:

$$\theta_d = -\frac{1}{2}\pi - \theta_m. \tag{14}$$

Again, inserting the lengths for the phalanges of each finger, yields the following values for θ_p and θ_d:

$$\begin{aligned}
\theta_{pi} &= \arcsin\left(\frac{-18+15.8}{22.4}\right) \approx -0.098 \quad (-5.64^\circ) \\
\theta_{pi} &= \arcsin\left(\frac{-18+17.4}{26.3}\right) \approx -0.023 \quad (-1.31^\circ) \\
\theta_{pi} &= \arcsin\left(\frac{-18+17.3}{25.7}\right) \approx -0.027 \quad (-1.56^\circ) \\
\theta_{pi} &= \arcsin\left(\frac{-18+16.0}{18.1}\right) \approx -0.027 \quad (-1.56^\circ)
\end{aligned}, \tag{15}$$

and

$$\begin{aligned}
\theta_{di} &= -\frac{1}{2}\pi - (-0.098) \approx -1.47 \quad (-84.36^\circ) \\
\theta_{dm} &= -\frac{1}{2}\pi - (-0.023) \approx -1.55 \quad (-88.69^\circ) \\
\theta_{dr} &= -\frac{1}{2}\pi - (-0.027) \approx -1.54 \quad (-88.44^\circ) \\
\theta_{dl} &= -\frac{1}{2}\pi - (-0.027) \approx -1.47 \quad (-84.36^\circ)
\end{aligned}. \tag{16}$$

Fig. 7, shows the configurations of the fingers after the numerical values for the angles above have been inserted into the equations of the forward kinematics.

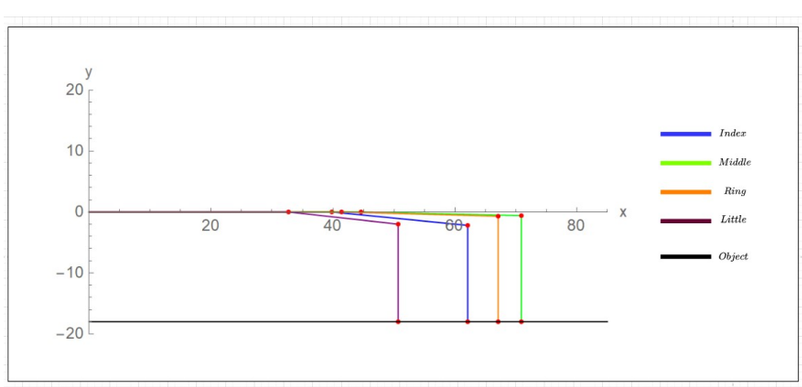


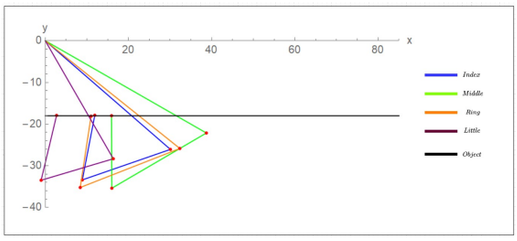
Figure 7: Fingertips placed at the object, while the proximal phalanges are horizontal and the distal phalanges are vertical.

2.1.5 Declined and Inclined Finger

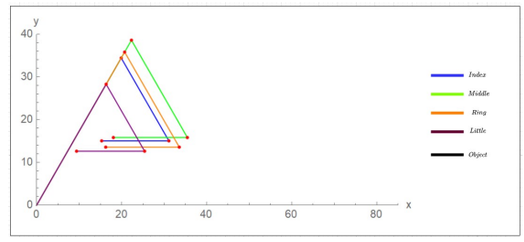
In this section, it will first be tried to mimic a declining finger that touches the boundary of the object from the bottom. However, the geometry involved in this problem makes it less straightforward to find an analytical solution for the angles. That is why the necessary joint angles are found by trial and error. A set of joint angles was found that brings the fingertip satisfactorily close to the object at $\theta_{mi} \approx -0.714$, $\theta_{mm} \approx -0.515$, $\theta_{mr} = -0.664$, $\theta_{pl} \approx -1.805$, and when all other angles were set to their minimum allowable value.

Additionally, another interesting case would be where the fingers are inclined. This would mimic a real-life situation where a human is grabbing onto an object. A similar situation is obtained when setting the

values for θ_m to their maximum, while the values of θ_p and θ_d are set to their minimum. Fig. 8 shows the resulting configurations of the fingers after the above conditions for the inclined and declined fingers are applied.



(a) Declined fingers that touch the object from the bottom



(b) Inclined fingers; Representing a situation where a human would grab onto an object.

Figure 8: Declined and inclined human fingers

2.2 Constrained Joint Angles

For real human fingers, it is not true that the joint angles can move entirely independently from each other. To get a more realistic kinematic model of the fingers, it is amused that the joint angle of the distal interphalangeal is a passive actuator that is controlled by the proximal interphalangeal joint angle (or vice versa) using the equation

$$\theta_d = \frac{2}{3}\theta_p. \tag{17}$$

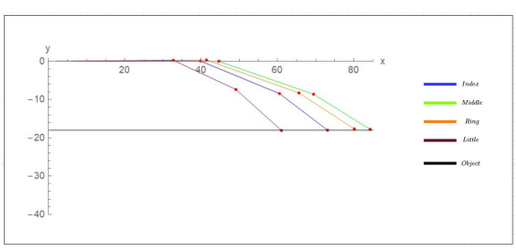
The metacarpophalangeal joint will be assumed to still move independently from the other joints. As a consequence, the forward kinematics of Eq. 2 and Eq. 3 will now be changed to

$$\begin{aligned} x &= p\cos(\theta_m) + i\cos(\theta_m + \theta_p) + d\cos\left(\theta_m + \theta_p + \frac{2}{3}\theta_p\right) \\ &= p\cos(\theta_m) + i\cos(\theta_m + \theta_p) + d\cos\left(\theta_m + \frac{5}{3}\theta_p\right) \end{aligned} \tag{18}$$

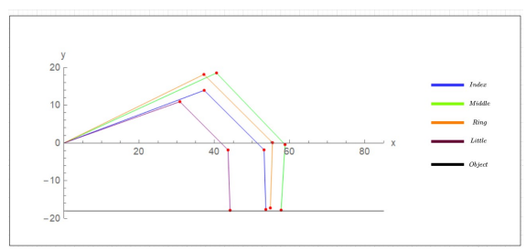
and

$$\begin{aligned} y &= p\sin(\theta_m) + i\sin(\theta_m + \theta_p) + d\sin\left(\theta_m + \theta_p + \frac{2}{3}\theta_p\right) \\ &= p\sin(\theta_m) + i\sin(\theta_m + \theta_p) + d\sin\left(\theta_m + \frac{5}{3}\theta_p\right). \end{aligned} \tag{19}$$

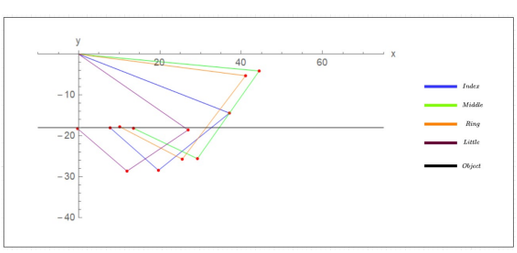
It should be noted that this dependency in the angles makes it harder to find the required joint angles to bring the fingertip to a target position when only the forward kinematics can be used. The geometry of the kinematic model will be harder to predict when only two joint angles can be inserted. Fig. 9 shows a number of configurations for the constrained model that are similar to the ones investigated for the model with three independent angles.



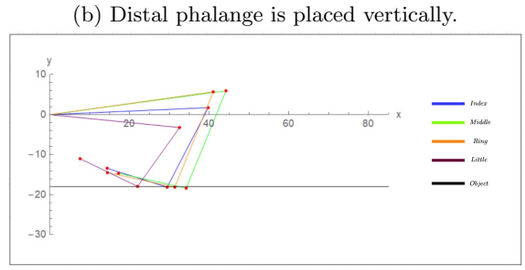
(a) Proximal phalange is placed horizontally.



(b) Distal phalange is placed vertically.



(c) Declined finger



(d) Distal interphalangeal joint in close contact with object

Figure 9: Interesting configurations that bring the fingertips close to the object.

The corresponding angles for these configurations have been obtained by trial and error and are shown in Tab. 2.

Although it is possible to get the fingertips close to the target position by changing the values of θ_m and θ_p, this is not ideal as these values have to be chosen manually and the solution does not guarantee that

Table 2: Values of θ_m and θ_p for the interesting configurations

Finger	θ_m	θ_p	**Finger**	θ_m	θ_p
Proximal Horizontal			**Distal Vertical**		
Index	0	-0.383 ($-21.94°$)	Index	0.354 (20.28°)	-1.146 ($-65.66°$)
Middle	0	-0.325 ($-18.62°$)	Middle	0.438 (25.10°)	-1.252 ($-71.73°$)
Ring	0	-0.341 ($-19.54°$)	Ring	0.431 (24.69°)	-1.217 ($-69.73°$)
Little	0	-0.438 ($-25.10°$)	Little	0.348 (19.94°)	-1.150 ($-65.89°$)
Declined Finger			**Inclined Finger**		
Index	-0.371 ($-21.26°$)	-2.094 ($-120°$)	Index	0.0434 (2.49°)	-2.094 ($-120°$)
Middle	-0.0922 ($-5.28°$)	-2.094 ($-120°$)	Middle	0.134 (7.68°)	-2.094 ($-120°$)
Ring	-0.128 ($-7.33°$)	-2.094 ($-120°$)	Ring	0.138 (7.91°)	-2.094 ($-120°$)
Little	-0.603 ($-34.55°$)	-1.948 ($-111.61°$)	Little	-0.0984 ($-5.64°$)	-2.094 ($-120°$)

the fingertip is actually at the desired position. Therefore, the inverse kinematics problem should be solved which will be discussed in the upcoming sections.

3 Jacobian Iterative Solver

Ideally, there are enough geometrical relations within the kinematic model given to derive analytical solutions for the inverse kinematics problem, which determine the values of the joint angles when the x and y positions are the input values. However, in the current kinematic model, not enough of these relations are given, and thus no analytical solution can be derived. A way around this problem is to use numerical iterative methods. These methods take an initial estimate for the joint configurations and iteratively refine these values until the position of the end-effector (in this case, the fingertip) is at a position that is close enough to the desired position. The mathematical form of this iterative solver is

$$q^{i+1} = q^i + J^{-1} q^i \delta T(q^i), \tag{20}$$

where q^i is the vector containing the current estimated values of the joint angles, $J^{-1}(q^i)$ is the inverse Jacobian matrix as a function of q^i, $\delta T(q^i)$ is the error of the current guess compared to the target position, and q^{i+1} is the vector containing the updated values of the joint angles. In the case of the constrained joint angles, the Jacobian matrix is of the form

$$J = \begin{pmatrix} \frac{\partial x}{\partial \theta_m} & \frac{\partial x}{\partial \theta_p} \\ \frac{\partial y}{\partial \theta_m} & \frac{\partial y}{\partial \theta_p} \end{pmatrix}, \tag{21}$$

where the partial derivatives are as follows:

$$\begin{aligned} \frac{\partial x}{\partial \theta_m} &= -m \sin(\theta_m) - p \sin(\theta_m + \theta_p) - d \sin\left(\theta_m + \frac{5}{3}\theta_p\right) \\ \frac{\partial x}{\partial \theta_p} &= -p \sin(\theta_m + \theta_p) - \frac{5}{3} d \sin\left(\theta_m + \frac{5}{3}\theta_p\right) \\ \frac{\partial y}{\partial \theta_m} &= m \cos(\theta_m) + p \cos(\theta_m + \theta_p) + d \cos\left(\theta_m + \frac{5}{3}\theta_p\right) \\ \frac{\partial y}{\partial \theta_p} &= p \cos(\theta_m + \theta_p) + \frac{5}{3} p \cos\left(\theta_m + \frac{5}{3}\theta_p\right) \end{aligned} \tag{22}$$

In this case, the Jacobian is a 2x2 matrix and its inverse is thus

$$J^{-1}(q^i) = \frac{1}{\left(\frac{\partial x}{\partial \theta_m}\right) \cdot \left(\frac{\partial y}{\partial \theta_p}\right) - \left(\frac{\partial y}{\partial \theta_m}\right) \cdot \left(\frac{\partial x}{\partial \theta_p}\right)} \begin{pmatrix} \frac{\partial y}{\partial \theta_p} & -\frac{\partial x}{\partial \theta_m} \\ -\frac{\partial y}{\partial \theta_p} & \frac{\partial x}{\partial \theta_m} \end{pmatrix}. \tag{23}$$

Lastly, the error vector is mathematically defined as

$$\delta T(q^i) = \begin{pmatrix} x_t \\ y_t \end{pmatrix} - \begin{pmatrix} x^i \\ y^i \end{pmatrix} = \begin{pmatrix} x_t \\ y_t \end{pmatrix} - \begin{pmatrix} p \cos\left(\theta_m^i\right) + i \cos\left(\theta_m^i + \theta_p^i\right) + d \cos\left(\theta_m^i + \frac{5}{3}\theta_p^i\right) \\ p \sin\left(\theta_m^i\right) + i \sin\left(\theta_m^i + \theta_p^i\right) + d \sin\left(\theta_m^i + \frac{5}{3}\theta_p^i\right) \end{pmatrix}, \tag{24}$$

where (x_t, y_t) are the inserted target values for the x and y position. In the coming sections, the performance of this iterative method will be evaluated. Furthermore, it will be shown that this method has some limitations and a couple of modifications are suggested along with their performances.

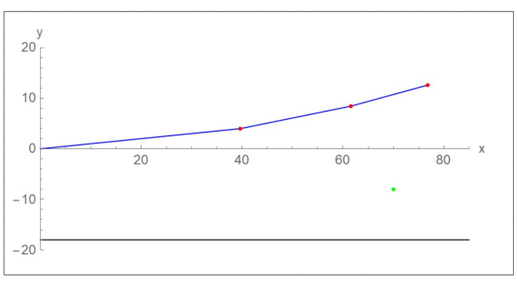
3.1 Standard Version

Firstly, the performance of the standard Jacobian inverse method will be evaluated. Alg. 1 shows how to apply this method using pseudo-code. Fig. 10 shows the performance of this algorithm for a set of different target positions.

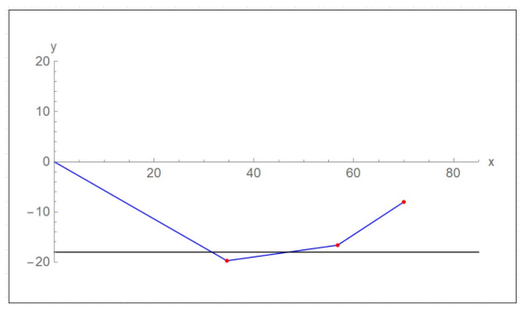
The method seems to be working rather well in these examples, as it found values for θ_m and θ_p within ten iterations that would bring the fingertip within a 0.005 distance of the target position. However, Fig. 10b, 10c, and 10d show a common problem with this solver method: the finger is placed in a position that is impossible to reach because of the boundary values of the angles. Furthermore, this method also does not take the presence of the object into account. Although this problem could perhaps be solved by choosing a a better set of values for the initial guess, it would be desired if the Jacobian iterative method could be modified in such a way that takes these two factors into account. In the next sections, some modifications are proposed that could help resolve this problem.

Algorithm 1 Application of inverse Jacobian iterative method.

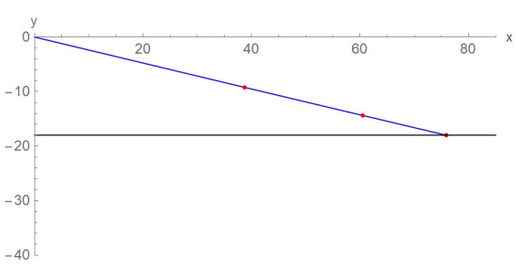
```
{p, i, d} = {39.8, 22.4, 15.8}                    ▷ Index finger phalanges.
t ← (x_t, y_t)^T   ▷ Where x_t, y_t are the x and y coordinates of the target position respectively.
q ← (θ_m, θ_p)^T   ▷ Here {θ_m, θ_p} are ({θ_m^0, θ_p^0}.
function J^{-1}(θ_m, θ_p)
    return [ ∂x/∂θ_p   ∂x/∂θ_m ; ∂y/∂θ_p   ∂y/∂θ_m ]^{-1}
end function                                      ▷ Inverse Jacobian matrix
results ← {{θ_m, θ_p}}                            ▷ Array for storing results. First entry: θ_m^0, θ_p^0
Dist ← 1                                          ▷ Distance between the target and the current position.
ii ← 0                                            ▷ This denotes the number of iterations.
while dist ≤ ϵ do                                 ▷ ϵ is maximum allowable error
    ii++
    x_c ← p cos(θ_m) + i cos(θ_m + θ_p) + d cos(θ_m + 5/3 θ_p)
    y_c ← p sin(θ_m) + i sin(θ_m + θ_p) + d sin(θ_m + 5/3 θ_p)

    c ← (x_c, y_c)^T                              ▷ Current location of finger.
    dist ← sqrt((x_t − x_c)^2 + (y_t − y_c^2))    ▷ Euclidean distance: target and current position.
    δT ← t − c                                    ▷ Error
    q ← q + J^{-1}(θ_m, θ_p) δT                   ▷ This updates the joint angles.
    results[ii] ← {θ_m, θ_p}                      ▷ Storing the resulting angles for the ith iteration.
end while
```



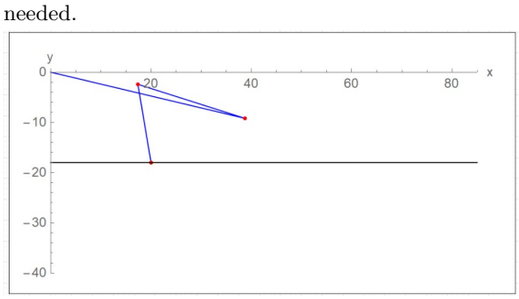
(a) Initial configuration of the index finger



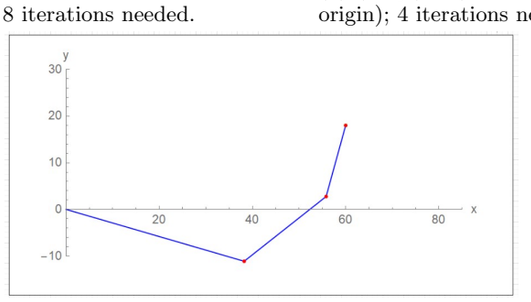
(b) $(x_t, y_t) = (70, -8)$ (point reasonably far away from origin but within reachable space); 5 iterations needed.



(c) $(x_t, y_t) = (75.8954, -18)$ (point furthest away on object within reachable space); 8 iterations needed.



(d) $(x_t, y_t) = (20, -18)$ (point relatively close to the origin); 4 iterations needed.



(e) $(x_t, y_t) = (60, 18)$ 4 iterations needed

Figure 10: Performance of the inverse Jacobian iterative solver for various target positions.

$$\left(\theta_m^0, \theta_p^0\right) = (0.1, 0.1), \text{ and } \epsilon = 0.005$$

3.2 Clipping and Brownian Motion

To compensate for the problem of reaching values for the joint angles which are outside of the boundary values, the clipping function is introduced. This function checks whether or not the values of θ_m and θ_p are

outside of their specified ranges after each iteration. If so, the values for these angles will be set back to the closest maximum allowable value. However, it was found in some cases that the method would get stuck on these boundary values, as it would go outside of the reachable space again right after it got clipped. To fix this problem, Brownian random motion was introduced. This was done by adding a random number to the values of the joint angles if the configuration got stuck. To evaluate when the solution is stuck, the Euclidean distance between the old and new values of the joint angles was computed, and if this value was less or equal to ϵ for three or more iterations, the configuration was said to be stuck. Alg. 2 shows the pseudo code of how this method was implemented.

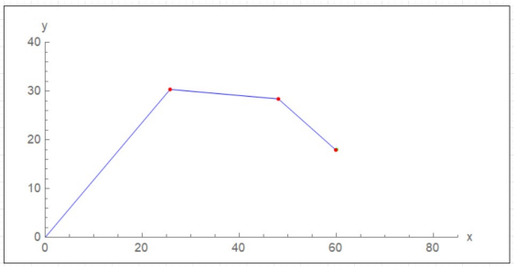
Algorithm 2 Application of inverse Jacobian iterative method in combination with Clipping and Brownian Motion.

$\{p, i, d\} = \{39.8, 22.4, 15.8\}$ ▷ Index finger phalanges.
$t \leftarrow \begin{pmatrix} x_t \\ y_t \end{pmatrix}$ ▷ Where x_t, y_t are the x and y coordinates of the target position respectively.
$q \leftarrow \begin{pmatrix} \theta_m \\ \theta_p \end{pmatrix}$ ▷ Here $\{\theta_m, \theta_p\}$ are $(\{\theta_m^0, \theta_p^0\}$.
function $J^{-1}(\theta_m,\theta_p)$
    return $\begin{pmatrix} \frac{\partial x}{\partial \theta_p} & \frac{\partial x}{\partial \theta_m} \\ \frac{\partial y}{\partial \theta_p} & \frac{\partial y}{\partial \theta_m} \end{pmatrix}^{-1}$
end function
function $Clip(\theta, low, high)$
$\theta \leftarrow \begin{cases} low & \theta < low \\ high & \theta > high \\ \theta & low < \theta < high \end{cases}$
    return θ
end function ▷ Inverse Jacobian matrix
$results \leftarrow \{\{\theta_m, \theta_p\}\}$ ▷ Array for storing results. First entry: θ_m^0, θ_p^0
$Dist \leftarrow 1$ ▷ Distance between the target and the current position.
$ii \leftarrow 0$ ▷ This denotes the number of iterations.
$nm \leftarrow 0$ ▷ Number of times the finger has not moved.
while $dist \leq \epsilon$ **do** ▷ ϵ is maximum allowable error
    $qprev \leftarrow q$ ▷ This stores the un-updated (previous) version of q
    $ii++$
    $x_c \leftarrow p\cos(\theta_m) + i\cos(\theta_m + \theta_p) + d\cos\left(\theta_m + \frac{5}{3}\theta_p\right)$
    $y_c \leftarrow p\sin(\theta_m) + i\sin(\theta_m + \theta_p) + d\sin\left(\theta_m + \frac{5}{3}\theta_p\right)$
    $c \leftarrow \begin{pmatrix} x_c \\ y_c \end{pmatrix}$ ▷ Current location of finger.
    $dist \leftarrow \sqrt{(x_t - x_c)^2 + (y_t - y_c^2)}$ ▷ Euclidean distance: target and current position.
    $\delta T \leftarrow t - c$ ▷ Error
    $q \leftarrow q + J^{-1}(\theta_m, \theta_p)\,\delta T$ ▷ This updates the joint angles.
    $Clip(\theta_m, -\frac{\pi}{3}, \frac{\pi}{3})$
    $Clip(\theta_p, -\frac{2\pi}{3}, 0)$ ▷ Clipping the values for θ_m and θ_p to their boundary values.
    $dist_q \leftarrow \sqrt{(qprev_{|\theta_m} - q_{|\theta_m})^2 + (qprev_{|\theta_p} - q_{|\theta_p})^2}$ ▷ distance of joint angles before and after iteration.
    if $dist_q \leq \epsilon$ **then**
        $nm++$
    end if
    if $nm \geq 0$ **then** ▷ This implements the Brownian motion.
        $\theta_m \leftarrow \theta_m + RandomReal(-\frac{\pi}{2}, \frac{\pi}{2})$ ▷ $RandomReal()$ gives random number within range (a, b).
        $\theta_p \leftarrow \theta_p + RandomReal(-\frac{\pi}{2}, \frac{\pi}{2})$
        $q \leftarrow \begin{pmatrix} \theta_m \\ \theta_p \end{pmatrix}$

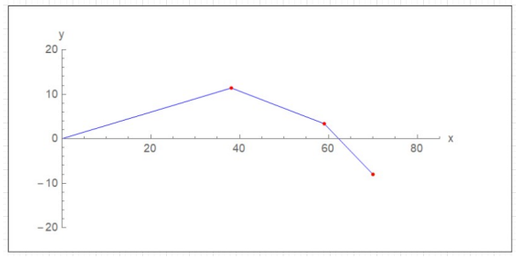
        $nm \leftarrow 0$
    end if
    $results[ii] \leftarrow \{\theta_m, \theta_p\}$ ▷ Storing the resulting angles for the ith iteration.
end while

Fig. 11 shows that for some target positions, such as $(x_t, y_t) = \{(60, 8), (70, -8), (20, -18)\}$, the clipping method outperforms the standard inverse Jacobian iterative method, since the first method needed a less or equal number of iterations and found a solution that was acceptable in terms of the boundaries of the angles. However, when the target position was set at $(x_t, y_t) = (50, 18)$, the clipping method got stuck even when the Brownian motion was applied (see Fig. 11d). Another example is when the target position was set at the point furthest away along the object which is still reachable. The clipping method did not get stuck, but it did not find a solution either even after more than 10000 iterations and using Brownian motion. In this case, the standard Jacobian method seems to outperform the clipping method, as it found the solution after 8 iterations.

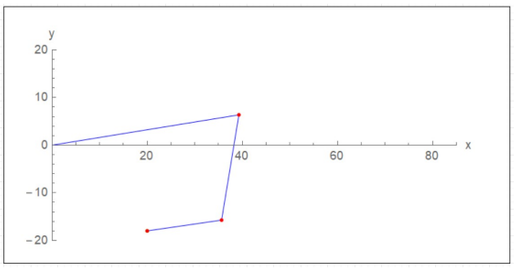
In general, it can be said that the use of Brownian motion does not seem to contribute to the solver method. In the cases when the finger got stuck and Brownian motion was applied, the finger did not find a solution even after more than thousands of iterations.



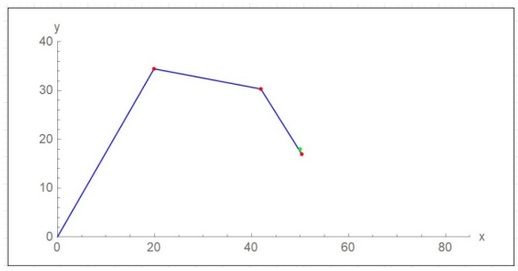
(a) $(x_t, y_t) = (60, 18)$; 5 iterations needed.



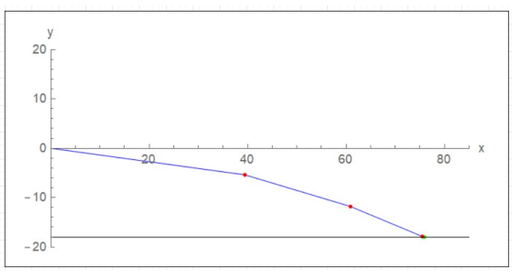
(b) $(x_t, y_t) = (70, -8)$; 5 iterations needed.



(c) $(x_t, y_t) = (20, -18)$; 5 iterations needed.



(d) $(x_t, y_t) = (50, 18)$; Solver got stuck even after > 10000 iterations



(e) $(x_t, y_t) = (75.8954, -18)$ (point furthest away on the object within reachable space); No solution found after > 10000 iterations.

Figure 11: Performance of the inverse Jacobian iterative solver in combination with clipping and Brownian motion for various target positions.

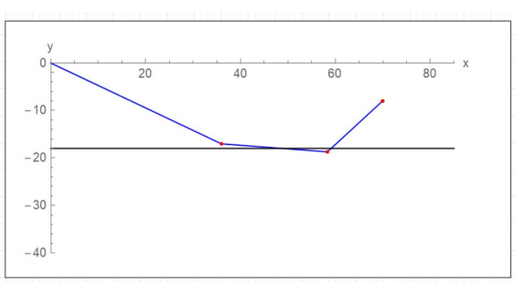
$$\left(\theta_m^0, \theta_p^0\right) = (0.1, 0.1) \text{ and } \epsilon = 0.005.$$

3.3 Psuedo-Inverse

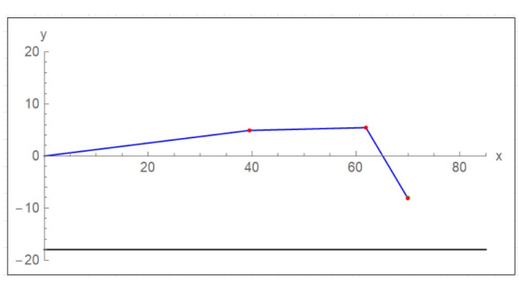
In the case of independent joint angles, the number of joint variables differs from the number of configuration parameters. There are three joint angles, while a two-dimensional Euclidean workspace is considered, meaning that there are two configuration parameters. Therefore, the current model is an under-determined system, meaning that there are infinitely many solutions for the joint angles of a given target position. Another problem is that the Jacobian matrix will now be rectangular, as it has more columns than rows, and as a consequence, the inverse Jacobian is not defined. A solution to this problem is to use the Moore-Penrose pseudo-inverse. These pseudo inverses give the set of solutions with the minimum norm (in case of an under-determined system) or the least squares solution (in case of an over-determined system). The pseudo-inverse for the minimum norm solution, the right inverse, is given as

$$A^+ = A^T \left(AA^T\right)^{-1}, \tag{25}$$

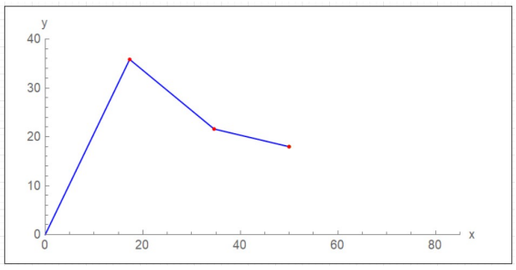
where A is a nxm matrix with $m > n$. Inserting Eq. 25 into Alg. 1 or 2, and altering the equations for the forward kinematics, will yield an algorithm for computing the joint angles of the kinematic model with three independent angles. Fig. 12 shows the resulting configurations for a number of different inserted target positions for both a pseudo-inverse model with and without clipped values. Note that the Brownian motion was not needed in these examples, as the solver did not get stuck on a certain set of values.



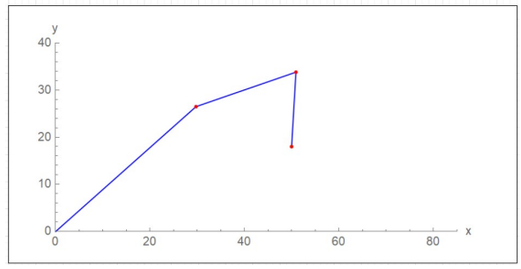
(a) Pseudo-inverse; $(x_t, y_t) = (70, -8)$; 5 iterations needed.



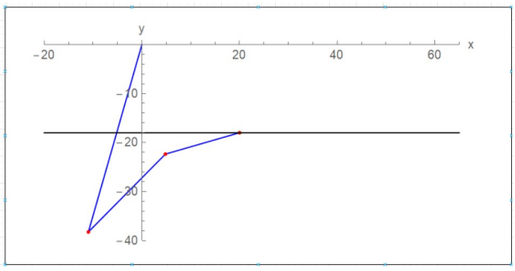
(b) Pseudo-inverse with clipping; $(x_t, y_t) = (70, -8)$; 12 iterations needed.



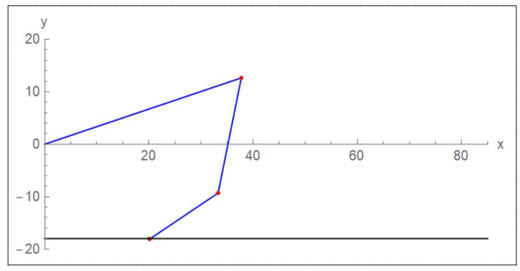
(c) Pseudo-inverse; $(x_t, y_t) = (50, 18)$; 7 iterations needed.



(d) Pseudo-inverse with clipping; $(x_t, y_t) = (50, 18)$; 5 iterations needed



(e) Pseudo-inverse; $(x_t, y_t) = (20, -18)$; 8 iterations needed.



(f) Pseudo-inverse with clipping; $(x_t, y_t) = (20, -18)$; 5 iterations needed.

Figure 12: Performance of the pseudo-inverse Jacobian iterative solver with and without clipping for various target positions.

$$\left(\theta_m^0, \theta_p^0, \theta_d^0\right) = (0.1, 0.1, 0.1) \text{ and } \epsilon = 0.005.$$

The results show that the pseudo-inverse method works rather satisfactorily, as it always found a solution within 10 iterations. However, these solutions were all not within the boundary values of the angles. The clipped version did find an acceptable solution in all cases for the joint angles, but it usually had to take a few more iterations.

4 Conclusion and Discussion

This project report has shown the full design of the kinematic model for human fingers. Starting with explaining the different parts of the hand and fingers and how this structure can be transformed into a kinematic model. From there onwards, the forward kinematics were derived and a couple of interesting examples of the model were shown in order to better understand its limitations. First, a model with three independent joint angles was considered and after that, a dependency between two joint angles was introduced in order to get a more realistic model of the finger.

It was stated that if the x and y position of the fingertip are desired to be the input values, the inverse kinematics problem needs to be solved as this would yield a function of the actuators (the angles) in terms of these input values. However, not enough geometrical relations within the model are given to come to an analytical solution. Therefore numerical and iterative methods were introduced, which can approximate the necessary values for the joint angles. Firstly, the standard inverse Jacobian iterative method was used. This method usually found a solution within 10 iterations, but these solutions were unacceptable because they require the joint angles to be outside of their boundary values. To solve this problem, the clipping function was introduced.

To solve this problem, the clipping function was introduced. This function checks after each iteration whether or not the values of the angles are outside of their boundaries. If so, the value of the angles will be set to the closest boundary value, and the next iteration will take place. It turns out that this function actually improves the solver methods, as it usually yields an acceptable solution. The only downside of this method is that it requires more iterations to get to a solution that is close enough to the target compared to the standard Jacobian inverse method. Furthermore, including the clipping function can cause the method to get stuck in one configuration, such as in the case where the target is the point furthest along the object within the reachable space.

To take this issue into account, Brownian random motion was introduced. I.e., if the solver did not change the values for the joint angles for a number of iterations, a random number was added to the angles in order to get it out of the position. However, there was not found a single example in this project where the use of Brownian motion actually helped to come to a solution.

Lastly, it was found that the inverse Jacobian solver method for a model with unconstrained joint angles can be used as well, even though the number of configuration parameters and joint angles is not equal. The right pseudo-inverse gives the minimum norm solution for an under-determined system, such as the human fingers with three independent angles. The Jacobian solver with the pseudo-inverse seems to be working rather satisfactorily, except for the fact that there is a substantial probability that it returns joint

angles that are not acceptable because of their boundary values. If the pseudo-inverse method is combined with the clipping function, the performance of the method increased substantially, looking at the examples investigated.

In general, the use of the clipping function has potential but the problem of the finger getting stuck in one configuration has to be resolved. It would perhaps still be interesting to investigate more into the use of Brownian motion, e.g. using a direction vector for the random motion (positive or negative) based on the sign and magnitude of the difference in current and target position. Lastly, it should be evaluated how the initial estimated configuration can be chosen in such a way that it yields a higher probability for the solver to end in an acceptable configuration.

References

[1] B. Alexander and V. Kotiuk, "Proportions of Hand Segments," *International Journal of Morphology*, vol. 28, pp. 755–758, Sep. 2010.